

THE WATERMARK THEOREM — LEAN CERTIFICATE

A machine-checked proof, in Lean 4 with Mathlib, that the lattice of the lines on the Fermat surface of prime degree $m \geq 5$ has discriminant $m^{3(m-3)^2}$

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Certificate v2 · 30 September 2026. v2 adds §10 (what was known before, from the sources read in the original), states Shioda's 1987 question exactly, and points to the Double Ladder certificate. The Lean content, the numbers and the fingerprints of v1 are unchanged.

Companion to *The Watermark Theorem* (R. Amichis Luengo, 4 June 2026), file `hodge-fermat-campaign/THE_WATERMARK_THEOREM.md` of the repository github.com/tretoef-estrella/chaise-longue-theorem, and papers/`THE_WATERMARK_THEOREM.md` of github.com/tretoef-estrella/watermark-theorem. The Lean project and all the evidence are in the second repository, folder `lean/`.

0. Verdict

For every prime $m \geq 5$, Lean 4 proves the following about the lattice V spanned by the $3m^2$ lines of the Fermat surface of degree m , with the bilinear form given by their intersection numbers:

- V is a **free abelian group of rank $3(m-1)(m-2) + 1$** ;
- for **every** $\mathbb{Z}$ -basis b of V , the determinant of the Gram matrix of b is **exactly $m^{3(m-3)^2}$** , with sign $+$.

The paper states $|\text{disc } V| = m^{3(m-3)^2}$ and gives the sign separately. Lean proves the value with its sign. The determinant does not depend on the basis.

For prime m , Shioda asked in 1987 whether $|\det NS| = m^{3(m-3)^2}$ ([Sh87], Questions 7.2 and 7.4, p. 133); [SSvL, §4] calls the formula his conjecture. Schütt, Shioda and van Luijk checked it by computer for every odd $m \leq 81$ [SSvL]. Aljovin, Movasati and Villaflor listed, by computer, the elementary divisors of this lattice for $m \leq 14$ [AMV, Table 1]. In 2015 Shioda still used the formula inside a conjecture [Sh15, Conjecture 23]. As far as we know no proof existed before June 2026. §10 gives the sources, read in the original.

The same Lean project also proves the **Double Ladder Theorem**, the structure of the discriminant group V^*/V , on top of this one; see its certificate [DL-Lean].

The whole proof compiles on the author's Mac. It contains no `sorry`, no `admit`, no `native_decide`, no `decide` and no added axiom. The final theorem depends only on the three standard axioms of Lean and Mathlib: `propext`, `Classical.choice` and `Quot.sound`.

What Lean certifies, and what it does not. Lean works with an explicit integer matrix G of size $3m^2 \times 3m^2$ (§1.2). That G is the matrix of intersection numbers of the lines on the complex Fermat surface is geometry. It is proved by hand in §1.4, by solving linear equations, and checked by computer for $m = 5, 7, 11$. It is not formalized. Two further facts are cited, not formalized: that $A/\ker G$ is the lattice of lines inside $H^2(S_m, \mathbb{Z})$, and that for prime m this lattice is the whole Néron-Severi group [SSvL, Deg]. So:

Lean certifies the lattice statement completely. The statement about the Fermat surface follows from it by the identification of §1.4, proved by hand and checked by computer, and by the cited results of [SSvL] and [Deg].

§1 states exactly what is proved. §2 states exactly what is not. §3–§8 describe how the result was obtained and how to check it. §9 says in one paragraph what is claimed; §10 records what was known before.

1. The certified statement

1.1 The theorem, verbatim

Printed by Lean with #check (logs check_run7.log and checkfinal.log):

```
@watermark_theorem : ∀ {m : ℕ} [inst : Fact (Nat.Prime m)],
  5 ≤ m →
    Module.Free ℤ (V m) ∧
      Module.finiteRank ℤ (V m) = 3 * (m - 1) * (m - 2) + 1 ∧
        ∀ {ι : Type u_1} [inst_1 : Fintype ι] [inst_2 : DecidableEq ι] (b : Module.Basis ι ℤ (V m)),
          ((LinearMap.BilinForm.toMatrix b) (formV m)).det = 1m ^ (3 * (m - 3) ^ 2)
```

```
'Watermark.watermark_theorem' depends on axioms: [propext, Classical.choice, Quot.sound]
```

The basis b may be indexed by any finite type in any universe.

1.2 The definitions it uses, verbatim

These are all the definitions needed to read the statement. They are printed by #print in checkfinal.log. All of them are in the file RequestProject/Watermark/Defs.lean (188 lines), written in the first piece and never changed afterwards.

```

@[reducible] def Watermark.Point : ℕ → Type :=
fun m => ZMod m × ZMod m

@[reducible] def Watermark.Line : ℕ → Type :=
fun m => Fin 3 × Point m

def Watermark.gramUpper : (m : ℕ) → Line m → Line m → ℤ :=
fun m p q =>
  if p.1 = q.1 then if p.2 = q.2 then 2 - ↑m else if p.2.1 = q.2.1 ∨ p.2.2 = q.2.2 then 1 else 0
  else
    if p.1 = 0 ∧ q.1 = 1 then if p.2.1 - p.2.2 = q.2.1 - q.2.2 then 1 else 0
    else
      if p.1 = 1 ∧ q.1 = 2 then if p.2.1 + p.2.2 = q.2.1 + q.2.2 then 1 else 0
      else if p.1 = 0 ∧ q.1 = 2 then if q.2.1 - q.2.2 = p.2.1 + p.2.2 + 1 then 1 else 0 else 0

def Watermark.gram : (m : ℕ) → Matrix (Line m) (Line m) ℤ :=
fun m => Matrix.of fun p q => if p.1 ≤ q.1 then gramUpper m p q else gramUpper m q p

@[reducible] def Watermark.A : ℕ → Type :=
fun m => Line m → ℤ

def Watermark.gramLin : (m : ℕ) → [NeZero m] → A m →1[ℤ] A m :=
fun m [NeZero m] => (gram m).mulVecLin

def Watermark.K : (m : ℕ) → [NeZero m] → Submodule ℤ (A m) :=
fun m [NeZero m] => (gramLin m).ker

@[reducible] def Watermark.V : (m : ℕ) → [NeZero m] → Type :=
fun m [NeZero m] => A m / K m

def Watermark.gramForm : (m : ℕ) → [NeZero m] → LinearMap.BilinForm ℤ (A m) :=
fun m [NeZero m] => (Matrix.toLinearMap₂' ℤ) (gram m)

def Watermark.gramFormAux : (m : ℕ) → [inst : NeZero m] → V m →1[ℤ] A m →1[ℤ] ℤ :=
fun m [NeZero m] => (K m).liftQ (gramForm m).flip ...

def Watermark.formV : (m : ℕ) → [inst : NeZero m] → LinearMap.BilinForm ℤ (V m) :=
fun m [NeZero m] => (K m).liftQ (gramFormAux m).flip ...

```

The two $\dots$ are proof terms: that gramForm vanishes when one argument lies in K , so that it descends to the quotient. Lean checks them; they carry no data. By the lemma formV_mk , $\text{formV } [x] [y] = x \cdot G \cdot y$ for all $x, y \in A$.

1.3 Faithfulness to the paper, line by line

The paper is *The Watermark Theorem*, v1.0 of 4 June 2026, §2.

Paper	Lean	Agreement
the $3m^2$ lines, three families of m^2 , each indexed by $(k, l) \in (Z/m)^2$	$\text{Line } m = \text{Fin } 3 \times \text{ZMod } m \times \text{ZMod } m$	Identical. Families 1, 2, 3 of the paper are $j = 0, 1, 2$.
self-intersection $2 - m$	$p = q$ gives $2 - \uparrow m$	Identical.
same family: 1 iff $k = k'$ or $l = l'$, for distinct lines	$p.2.1 = q.2.1 \vee p.2.2 = q.2.2$ gives 1, else 0	Identical.
family 1 $\times$ family 2: $k - l \equiv k' - l'$	case $p.1 = 0 \wedge q.1 = 1$	Identical.
family 2 $\times$ family 3: $k + l \equiv k' + l'$	case $p.1 = 1 \wedge q.1 = 2$	Identical.
family 1 $\times$ family 3: $k' - l' \equiv k + l + 1$, with (k', l') in family 3	case $p.1 = 0 \wedge q.1 = 2$: $q.2.1 - q.2.2 = p.2.1 + p.2.2 + 1$	Identical; q is the line of family 3.

Paper	Lean	Agreement
the lower blocks by symmetry	<i>gram</i> reads <i>gramUpper</i> with the arguments swapped when $p.1 > q.1$	Identical. <i>gram_transpose</i> proves $G^T = G$.
$A = \bigoplus_j \mathbb{Z}/m^2$, G the Gram matrix	$A\ m = \text{Line } m \rightarrow \mathbb{Z}, \text{gram } m$	Identical.
$V :=$ the lattice spanned by the classes of the lines	$V\ m = A\ m / \ker G$ with the form induced by G	The same lattice: see §1.4, point 3.
$m \geq 5$ prime	$[\text{Fact } (\text{Nat.Prime } m)], 5 \leq m$	Identical, and no other hypothesis.
$\text{rank } V = 3(m-1)(m-2) + 1$	$\text{Module.finerank } \mathbb{Z} (V\ m) = 3 * (m - 1) * (m - 2) + 1$	Identical. The subtraction in $\mathbb{N}$ never truncates, because $m \geq 5$.
$ \text{disc } V = m^{3(m-3)^2}$, and $\text{disc } V = (-1)^{\text{rank } V - 1} \text{disc } V $	$\det (\text{toMatrix } b (\text{formV } m)) = m ^ (3 * (m - 3) ^ 2)$ for every basis b	Lean proves the value with its sign. The sign + agrees with the paper, because $\text{rank } V - 1 = 3(m-1)(m-2)$ is even.

Non-vacuity.

- «Free of rank r » is stated as $\text{Module.Free } \mathbb{Z} (V\ m) \wedge \text{Module.finerank } \mathbb{Z} (V\ m) = r$. In Mathlib the rank of a module that is not finitely generated is 0. Here $r = 3(m-1)(m-2)+1 \geq 37$, so the statement forces $V\ m \cong \mathbb{Z}^r$.
- The clause «for every basis» is therefore not empty. This is also checked in Lean on an instance: the file `checks/CheckFinal.lean` proves, from the theorem, that at $m = 5$ a $\mathbb{Z}$ -basis of $V\ 5$ exists, has 37 elements, and has Gram determinant 5^{12} . It compiles with no error (`checkfinal.log`).
- The hypotheses can be satisfied: $m = 5$ is used in that check.

1.4 The matrix is the geometry

This part is **not** formalized. It connects the matrix G of §1.2 to the Fermat surface $S_m : x_0^m + x_1^m + x_2^m + x_3^m = 0$ in $P^3(C)$.

1. The lines. Put $\varepsilon = e^{\pi i/m}$ and $\zeta = e^{2\pi i/m}$, so that $\varepsilon^m = -1$. For $a, b \in \mathbb{Z}/m$, the following three kinds of line lie on S_m :

- family 1: $x_0 = \varepsilon \zeta^a x_1, x_2 = \varepsilon \zeta^b x_3$;
- family 2: $x_0 = \varepsilon \zeta^a x_2, x_1 = \varepsilon \zeta^b x_3$;
- family 3: $x_0 = \varepsilon \zeta^a x_3, x_1 = \varepsilon \zeta^b x_2$.

For example, on a line of family 1, $x_0^m + x_1^m = (\varepsilon^m + 1)x_1^m = 0$ and likewise $x_2^m + x_3^m = 0$. These are the $3m^2$ lines of S_m [SSvL]. The line with parameters (a, b) in family f corresponds to the Lean index $(f-1, (a, b))$.

2. The intersection numbers.

- A line has self-intersection $2 - m$: by adjunction, $L^2 = -2 - K \cdot L$ and $K = (m-4)H$.
- Two distinct lines meet transversally in at most one point, so their intersection number is 1 if they meet and 0 if not.
- They meet exactly when their four linear equations have a common non-zero solution. Solving the four equations gives:
 - same family: they meet iff they share a or share b ;
 - families 1 and 2, with parameters (a, b) and (c, d) : iff $\varepsilon \zeta^a \cdot \varepsilon \zeta^d = \varepsilon \zeta^b \cdot \varepsilon \zeta^c$, i.e. $a - b \equiv c - d$;
 - families 2 and 3, with (c, d) and (e, f) : iff $c + d \equiv e + f$;

- families 1 and 3, with (a, b) and (e, f) : from $x_0 = \varepsilon \zeta^a x_1 = \varepsilon \zeta^a \varepsilon \zeta^f x_2 = \varepsilon \zeta^a \varepsilon \zeta^f \varepsilon \zeta^b x_3 = \varepsilon \zeta^e x_3$ one gets $\varepsilon^2 \zeta^{a+b+f} = \zeta^e$, and $\varepsilon^2 = \zeta$. So they meet iff $e - f \equiv a + b + 1$.

These are exactly the six cases of `gramUpper`. The $+1$ of the last case is $\varepsilon^2 = \zeta$.

3. The lattice.

- $H^2(S_m, \mathbb{Z})$ is torsion free and its intersection form is unimodular.
- The classes of the lines span $NS(S_m) \otimes \mathbb{Q}$ if and only if $m \leq 4$ or $\gcd(m, 6) = 1$: a theorem of Shioda, recalled in [SSvL, §3]. This covers every prime $m \geq 5$.
- The intersection form on $NS(S_m)$ is non-degenerate (Hodge index).

So a combination $x \in A$ of lines has class 0 if and only if it pairs to zero with every line, that is, if and only if $Gx = 0$. Hence the lattice spanned by the lines, with its intersection form, is $A/\ker G$ with the form induced by G . This is V_m with `formV m`.

4. Computer check. The script `checks/wfinal_geometry.py` builds the three families of lines numerically and decides, for every pair, whether the four equations have a common non-zero solution (rank of a 4×4 complex matrix). It compares the result with a verbatim transcription of `gramUpper`.

- $m = 5, 7, 11$: all 2 850, 10 878 and 66 066 pairs agree.
- **Control**: the same comparison with $k + l - 1$ in place of $k + l + 1$ in the last case disagrees on 250, 686 and 2 662 pairs.

Log: `checks/wfinal_geometry.log`.

5. The Néron–Severi group. For $\gcd(m, 6) = 1$ the lines generate $NS(S_m)$ over $\mathbb{Z}$: [SSvL] for $m \leq 100$, and [Deg] for every such m . So for every prime $m \geq 5$, $V = NS(S_m)$ and the theorem gives $\text{disc } NS(S_m) = m^{3(m-3)^2}$. This is cited, not formalized.

1.5 The main intermediate theorems

Each is printed in the check log of its run, with axioms [propext, Classical.choice, Quot.sound].

- `SAT_le_K (W3)`: the explicit sublattice SAT lies in $\ker G$, **for every $m \geq 1$** . The paper records the corresponding membership (§7 there) as verified by computer only; Lean proves it for every m .
- `finrank_ker_satMap = 12` and `finrank_range_satMap = 9m - 19 (W2)`: Lemma C of the paper, the relation count modulo m .
- `finrank_SAT = 9m - 7` for odd m (W3), and `finrank_K = 9m - 7` for prime $m \geq 5$ (W5).
- `smul_mem_SAT (W5)`: $m \cdot x \in SAT$ for every $x \in \ker G$, with an explicit witness.
- `index_SAT (W5)`: $[K : SAT] = m^{12}$.
- `index_Lfam (W6)`: the lattice of pencil pullbacks on one family has index $m^{(m^2+m+2)/2}$ in $\mathbb{Z}^{m^2}$, for every prime m .
- `F_inf_K`, `finrank_F`, `index_F_sup_K (W6)`: the sublattice F meets $\ker G$ in 0, has rank $3(m-1)(m-2)+1$, and $[A : F + K] = m^{3(m^2+m+2)/2 - 12}$.
- `V_free (W7)`: V_m is free for **every $m \geq 1$** .

2. What is not certified

1. The geometry of §1.4.

- That G is the intersection matrix of the lines on S_m : proved by hand in §1.4 and checked by computer at $m = 5, 7, 11$, not formalized.
- That $A/\ker G$ is the lattice of lines in $H^2(S_m, \mathbb{Z})$: standard facts, cited.

Mathlib has no complex algebraic surfaces and no singular cohomology of them.

1. **The equality $V = NS(S_m)$** for prime m : [SSvL] and [Deg], cited. With it, the theorem computes $\text{disc } NS(S_m)$.

2. **The paper's own proof is not the proof formalized.**

- The paper proves the theorem through characters of $(\mathbb{Z}/m)^3$, trace lattices and a large determinant (§§3–10 there).
- Lean follows a different route, entirely over the integers and without characters (§3.1 below). It was designed for the formalization and checked numerically before each piece was sent.
- The Lean proof is therefore a second, independent proof of the same statement. It will be written as a paper in a later version.

3. **Not formalized, and not claimed here:**

- odd composite m and even m , where the formula is conjectural (paper §2);
- the corollaries of the paper's §12: the transcendental lattice and the Brauer group;
- *The Double Ladder Theorem* (the group structure of V^*/V).

4. **Faithfulness is an audit, not a theorem.**

- Lean guarantees that the proof is correct for the statement as written. That the statement says what the paper says is checked in §1.3.
- A reader should check §1.2–§1.3 independently: one short file, `Defs.lean`, and six cases of `gramUpper`. This is the one place where trust in the author remains, together with the geometry of §1.4.

5. **Timing data.** Aristotle's own start and end times per task were not exported. The times in §6 are the download times of the result files: an upper bound for completion, not a measure of Aristotle's work.

3. How the proof is organised in Lean

The proof was cut into seven pieces, W1–W7. Each piece is one self-contained Markdown file in `pieces/`, with `Setting`, `Statements`, `Proof` and `Checks`. Each piece produced one Lean file in `RequestProject/Watermark/`.

No piece ever modified a file of an earlier piece. Every returned project was compared file by file with the compiled project, and all earlier `.lean` files were byte-identical, as were `lakefile.toml`, `lean-toolchain` and `lake-manifest.json`.

#	Piece	Content	Lean file	Main Lean results
W1	w1_gram_basics.md	lines, G , A , $K = \ker G$, V , the induced form; $G^T = G$; $G \cdot N_j = m \cdot I$; R.1; R.2	<code>Defs.lean</code> , <code>W1.lean</code>	<code>gram_transpose</code> , <code>gram_mulVec_famVec</code> , <code>sum_E_pencil</code> , <code>aug_of_mem_K</code> , <code>formV</code> , <code>formV_isSymm</code>
W2	w2_counting_lemma.md	the sublattice SAT (nine sum-zero functions and two scalars); Lemma C: kernel 12, range $9m - 19$	<code>W2.lean</code>	<code>SAT</code> , <code>finrank_satSrc</code> , <code>exists_satSrc_lift</code> , <code>finrank_ker_satMap</code> , <code>finrank_range_satMap</code>

#	Piece	Content	Lean file	Main Lean results
W3	w3_sat_in_K.md	$SAT \subseteq K$ for every m ; injectivity and rank $9m - 7$ for odd m	W3.lean	gram_mulVec_satMap, SAT_le_K, satMap_injOn, finrank_SAT
W4	w4_pencil_calculus.md	pullbacks along the $m + 1$ pencils; G on pullbacks: 0 on axis pencils, $-m$ on live pencils, six overlap formulas	W4.lean	pb, fib, gram_mulVec_pb_live, the six gram_mulVec_pb_* overlap formulas, sum_pencil_mul_pencil, smul_ext_eq_sum_pb_fib
W5	w5_mK_in_SAT.md	the census of $\ker G$; $m \cdot K \subseteq SAT$; rank $K = 9m - 7$; $[K : SAT] = m^{12}$	W5.lean	fib_live_const, fib_overlap_*, smul_mem_SAT, finrank_K, index_SAT
W6	w6_index.md	the sublattice F , its Gram matrix, $F \cap K = 0$, $[Z^{m^2} : L_j] = m^{(m^2+m+2)/2}$, $F + SAT = L$, $[A : F + K]$	W6.lean	gramForm_pb_pb_of_ne, gramForm_pb_pb_self, gramForm_famVec_self, F_inf_K, finrank_F, index_Lfam, F_sup_SAT, index_F_sup_K
W7	w7_discriminant.md	V free of rank $3(m-1)(m-2)+1$; the Gram determinant of F ; change of basis	W7.lean	V_free, finrank_V, det_gramC, index_map_F, exponent_eq, watermark_theorem

3.1 The chain of the final theorem

This is how watermark_theorem is proved in Lean. Every step is a Lean proof. Write $pb_j^\pi(u)$ for the pullback of a function $u : Z/m \rightarrow Z$ to family j along the pencil π , and Z_0 for the functions with sum 0.

1. **Freeness and rank** (W7, V_free, finrank_V). $V = A/\ker G$ is torsion free: if $c \cdot x \in \ker G$ with $c \neq 0$, then $x \in \ker G$. It is finitely generated, hence free. Its rank is $3m^2 - \text{rank } K = 3m^2 - (9m - 7) = 3(m-1)(m-2) + 1$. The value $\text{rank } K = 9m - 7$ comes from W5.

2. **G on pullbacks** (W4). For $u \in Z_0$:

- $G \cdot pb^\pi(u) = 0$ on the two axis pencils;
- $G \cdot pb^\pi(u) = -m \cdot pb^\pi(u)$ on the $m - 3$ live pencils $t \notin \{0, \pm 1\}$;
- on the six pencils ± 1 (two per family), $G \cdot pb^\pi(u) = -m \cdot pb^\pi(u) + m \cdot pb^{\pi'}(u')$, with π' a pencil ± 1 of another family, $u' = u$ for the pairs of families (1,2) and (2,3), and $u(x - 1)$ or $u(x + 1)$ for the pair (1,3).

3. **The kernel and SAT** (W2, W3, W5).

- $SAT \subseteq K$ for every m (W3).
- Pairing $x \in K$ against pullbacks gives the census: the fibre sums of x along the live pencils are constant, and so are the differences along the overlap pencils (W5).
- From the census one builds an explicit admissible datum D with $\text{satMap}(D) = m \cdot x$, so $m \cdot K \subseteq SAT$ (W5).
- With Lemma C (W2: kernel 12 modulo m) and $K \cap m \cdot A = m \cdot K$ (K is saturated), this gives $[K : SAT] = m^{12}$ (W5).

4. **The sublattice F and its index** (W6).

- F is spanned by the blocks $pb_j^{t_i}(Z_0)$ for the $3(m - 3)$ live pencils, one member of each of the three overlap pairs, and N_0 .
- By step 2, distinct blocks are B -orthogonal, $B(pb \ u, pb \ v) = -m^2 \sum u(x)v(x)$ inside a block, and $B(N_0, N_0) = m^3$. Positivity of $\sum u^2$ then gives $F \cap K = 0$.
- $F + SAT = L$, where L is the lattice of vectors whose restriction to each family lies in $L_j = \text{span of } 1 \text{ and all pencil pullbacks}$. The standard Gram matrix of L_j is block diagonal, so $[Z^{m^2} : L_j] = m^{(m^2+m+2)/2}$.

- Since $F \cap K = 0$, $[F + K : F + SAT] = [K : SAT] = m^{12}$, and so $[A : F + K] = m^{3(m^2+m+2)/2 - 12}$.

5. The determinant (W7).

- The images in V of the spanning vectors of F are linearly independent. Their Gram matrix is block diagonal: $3(m - 2)$ blocks $-m^2(I + J)$ of size $m - 1$ and one block m^3 . Its determinant is $m^{3(m-2)(2m-1)+3}$, with sign $+$ because $m - 1$ is even.
- They span a sublattice of V of index $[A : F + K]$. By `Submodule.natAbs_det_basis_change` in `Mathlib` and the change-of-basis formula $Gram_c = P^T Gram_b P$: $\det Gram_b = m^{3(m-2)(2m-1)+3} / m^{2(3(m^2+m+2)/2 - 12)} = m^{3(m-3)^2}$.
- A basis indexed by any other finite type is reindexed; the determinant does not change.

3.2 The dependency cone

A short Lean program (`checks/DepsFinal.lean`, `log deps_watermark.log`) collects every declaration that `watermark_theorem` uses, directly or indirectly, and keeps those defined in the project.

- **Size: 410** declarations of the project, in **8** of its 9 files: `Defs` 36, `W1` 34, `W2` 81, `W3` 38, `W4` 24, `W5` 50, `W6` 103, `W7` 44. Every one of them is checked by the kernel as part of `watermark_theorem`.
- The only file not in the cone is `RequestProject/Main.lean`: 24 lines of options and no declaration, the default header of an Aristotle project. The Watermark files do not import it.

So every piece is on the path of the final theorem, as §3.1 says.

4. Trust base

- **What must be trusted:**
 - the Lean 4 kernel, `toolchain leanprover/lean4:v4.28.0`;
 - `Mathlib` at `commit 8f9d9cff6bd728b17a24e163c9402775d9e6a365` (tag `v4.28.0`, `lake-manifest.json`);
 - the three standard axioms `propext`, `Classical.choice`, `Quot.sound`;
 - the audit of the statement (§1.3) and the geometry of §1.4.
- **What need not be trusted:** Aristotle (Harmonic), the AI system that wrote the Lean proofs, and the author's brute-force checks. Every file Aristotle returned was compiled again on the author's machine, and the kernel re-checks every proof.
- **Forbidden constructs, searched as whole words in all 9 files:** none of `sorry`, `admit`, `axiom`, `native_decide`, `implemented_by`, `extern`, `unsafe`, `opaque`, `csimp`, `partial`, `macro`, `elab`, `syntax`, `notation`, `debug` occurs.
- **The tactic decide does not occur:** the word `decide` is in none of the 9 files.
- **Options.** The eight Watermark files set **no option at all**, so they run with Lean's default resource limits. The 24-line file `Main.lean`, which is not imported, sets resource limits, printing options and `autoImplicit false`; none of them affects soundness.
- **Axioms of every main theorem**, printed after each piece (`logs check_run1.log ... check_run7.log`): always `[propext, Classical.choice, Quot.sound]`.
- **Aristotle's own register** for the project lists 31 properties PROVED, 0 NEGATED, 0 in progress.

5. The procedure

The procedure is the one used for the Lean certificate of *The Chaise Longue Theorem* [CL-Lean]. For each of the seven pieces:

1. **Write the piece.** A self-contained Markdown file with Setting, Statements, Proof and Checks. It names the earlier Lean results it may reuse and forbids changing any existing file or statement.
2. **Brute-force check before sending.** A Python or Macaulay2 program checks the statements in small cases, with a negative control where possible (a deliberately wrong variant that must fail). The numbers are in the table of §6.
3. **Send to Aristotle** (project 884019b5-ec06-4ff7-8a7b-8d8a1f18d0a1), one task per piece.
4. **Download** the resulting project (a `.tar.gz`), after reloading the project page and confirming that the task is finished, and extract it into `run<N>/`.
5. **Diff** against the locally compiled project. Stop if any earlier `.lean` file changed. None ever did.
6. **Grep** the new file for the forbidden constructs of §4.
7. **Build locally:** `lake build RequestProject.Watermark.W<N>` inside a watchdog that stops the build above 3 GB of memory or 15 minutes. Log `build_run<N>.log`.
8. **Print** the main statements (`#check`) and their axioms (`#print axioms`). Log `check_run<N>.log`.
9. **Audit faithfulness:** compare the printed statements with the piece and with the paper. Accept generalizations; stop on any weakening.
10. **Record** the run in the state file `ESTADO_WATERMARK.md`.

Stop conditions, fixed in advance:

- any property NEGATED by Aristotle;
- a statement that is not faithful;
- a build failure;
- a sorry or an extra axiom;
- a change to an old file.

None occurred.

Incidents, recorded as they happened:

- **W2.** The proof sketch in the piece contained an error: in the lift of a sum-zero function it subtracted $(\sum y_0)/m$ where $\sum y_0$ was needed. Aristotle noticed it and proved the statement with the correct lift. The statement was unaffected.
- **W3.** The author predicted that $SAT \subseteq K$ would fail for even m . His own pre-send check refuted the prediction (it holds at $m = 4, 6$), and the piece was stated for every m . Injectivity does fail for even m , and that part is stated for odd m .
- **W4.** The first pre-send test of the overlap formulas had the direction of the translation wrong in the pair of families 1 and 3 (Lean indices 0 and 2), and reported False (`checks/w4_route.log`). A diagnosis at $m = 5$ found the right direction: $u(x - 1)$ in $G \cdot pb_0^{+1}$ and $u(x + 1)$ in $G \cdot pb_2^{-1}$. The piece was sent with it. The corrected formulas were re-checked on 30 September at $m = 5, 7, 11$ on a basis of Z_0 : all true, and the swapped direction fails (`checks/w4_formulas_recheck.log`).
- **W5.** One download was premature: tarball (4) contains the state after W4, taken before W5 had finished. It was not used. Since then the project page is reloaded and the finished task confirmed before every download.

Several pieces were proved in a more general form than asked, and each is noted in the state file:

- unused hypotheses dropped: $SAT \subseteq K$ for every m , V free for every $m \geq 1$, the index of L_j for every prime;
- the basis in the final theorem indexed by any finite type;
- in W2, Lemma C proved by mixed second differences instead of the paper's polynomial dictionary, with the same statement.

6. Data per piece

- **Tarball, md5:** the project returned by Aristotle, file 884019b5-...-aristotle (N).tar.gz in ~/Downloads, first 8 hex digits of its md5.
- **Downloaded:** local time the file was saved (CEST). This bounds the completion of the Aristotle task.
- **Build, peak memory:** the local `lake build` of the new file, measured by the watchdog: wall-clock seconds including `lake`'s start-up, and the largest resident memory it sampled.
- **Lines / theorems / definitions:** of the file, in the final project. Theorems are the lines starting with *theorem* or *lemma*; definitions those starting with *def*, *abbrev*, *structure*, *inductive*, *instance* or *class* (possibly after *noncomputable*, *private* or *protected*).
- **Checks:** the brute-force check made before sending, with its negative control.

#	Aristotle task	Tarball	md5	Downloaded	Build (s)	Peak (GB)	Lines	Thms	Defs	Checks before sending
W1	bfec8091	(base)	fe4ea007	29 Sep 21:54	178	2.33	188 + 293	15 + 21	17 + 1	$m = 5, 7, 11$: G symmetric, $G \cdot N_j = m \cdot \mathbb{I}$, $R.1$; fibre formulas for $G \cdot y$. $m = 5, 7$ (Macaulay2): $m \mid \text{aug}(x_j)$ on K , rank $V = 37, 91$, product of the non-zero elementary divisors of $G = 5^{12}, 7^{48}$. Control: a partial fibre functional is not divisible by m on K
W2	f501afa3	(1)	c10d5251	29 Sep 22:21	53	2.12	554	32	23	$m = 3, 5, 7, 11$: the translation $x \mapsto x - 1$ in the term shared by families 1 and 3 is the only one with $SAT \subseteq K$; controls (other shifts, swapped pencils) fail. $m = 5, 7, 11$: $\text{rank}_Q = 9m - 7$, rank mod $m = 9m - 19$, kernel 12. Relations of dimension 12 at $m = 5, 7, 11, 13$; $m = 3$ gives 11 (control)
W3	d004cb9e	(2)	8a6ed828	29 Sep 22:45	47	2.08	467	33	1	$m = 3, 4, 5, 6, 7, 9, 11, 15$: a basis and 20 random admissible data lie in K ; $\text{rank}_Q = 9m - 7$ for odd m . Control: a datum without sum zero is not in K

#	Aristotle task	Tarball	md5	Downloaded	Build (s)	Peak (GB)	Lines	Thms	Defs	Checks before sending
W4	b499b938	(3)	f1a a3d 2c	29 Sep 23:09	51	2.17	355	25	5	overlap formulas, diagnosed at $m = 5$ (see §5), re-checked at $m = 5, 7, 11$ on a basis of Z_0 , control fails; $[Z^{m^2} : L_j] = 5^{16}, 7^{29}, 11^{67}$; $ det $ of the Gram of $F = 5^{84}, 7^{198}$
W5	2aa54b1f	(5)	c91 68c 9c	29 Sep 23:37	76	1.64	413	30	6	$m = 5, 7$: census and $\mathfrak{m} \cdot \mathfrak{x} = \text{satMap}(D)$ with D admissible, exact on all 38 and 56 vectors of a basis of $\ker G$ (Macaulay2)
W6	cdf7f149	(6)	1eb 51b 50	30 Sep 00:08	64	1.81	802	61	18	$m = 5, 7$: F and SAT together have $3m^2$ generators and $[A : F + SAT] = 5^{48}, 7^{87}$ exactly; $m = 5$ (Macaulay2): $[A : F + K] = 5^{36}$, $[K : SAT] = 5^{12}$; $m = 7$: $[A : F + K] = 7^{75}$
W7	c097cb76	(7)	c0b 960 aa	30 Sep 07:04	89	1.37	249	15	5	$m = 5, 7$: det of the Gram of $F = +5^{84}, +7^{198}$ (sign checked); hence $det \text{Gram}_b = 5^{12}, 7^{48}$

Notes on the table.

- The build of W1 compiled `Main.lean` (161 s), `Defs.lean` and `W1.lean` (60 s) together. Its lines, theorems and definitions are given as $Defs + W1$.
- Tarball (4) (md5 c098ed1c, 29 Sep 23:24) is the premature download of §5, not used.
- At $m = 7$ the Macaulay2 computation of $[K : SAT]$ was stopped by the watchdog at 1.2 GB. It is not needed: $[K : SAT] = m^{12}$ is a Lean theorem.
- The builds are incremental: each builds only the new file on top of the compiled project. The clean rebuild of everything is in §8.
- The check runs (`#check` and `#print axioms`) took 92–139 s each, peak 1.59–2.52 GB (logs `check_run*.log`).

7. Totals

Quantity	Value
Pieces sent to Aristotle	7 (one task each)
Properties registered by Aristotle	31 PROVED, 0 NEGATED, 0 in progress
Lean files	9 (<i>Defs</i> , <i>W1–W7</i> , and the unused header <i>Main.lean</i>)
Lines of Lean	3 345
Theorems and lemmas (<i>theorem/lemma</i> at the start of a line)	232
Definitions (<i>def, abbrev, structure, inductive, instance, class</i>)	76

Quantity	Value
sorry / admit / added axioms / native_decide / decide	0 / 0 / 0 / 0 / 0
Axioms of watermark_theorem	propext, Classical.choice, Quot.sound
Calendar time	first result downloaded 29 Sep 21:54, last 30 Sep 07:04; work paused during the night between W7 being sent and its download
Sum of the 7 incremental local builds	558 s = 9.3 min
Largest local build	178 s (W1, with the project header); peak memory 2.33 GB (W1)
Clean rebuild of the whole project, one module at a time	9 of 9 modules, all exit 0; 07:43:16 → 07:54:21 on 30 Sep, i.e. 11 min 05 s wall-clock (658 s inside the 9 lake build calls); slowest module Main.lean, 168 s, and among the Watermark files W7, 89 s; peak memory 2.16 GB (Defs)
Declarations of the project in the dependency cone of watermark_theorem	410, in 8 files (§3.2)

8. Reproducibility

Machine: Apple Mac (arm64), 8 CPU cores, 8 GB of memory, macOS (Darwin 25.5.0).

Toolchain: leanprover/lean4:v4.28.0; Mathlib v4.28.0 = commit 8f9d9cff6b-d728b17a24e163c9402775d9e6a365 (from lake-manifest.json). These are the same toolchain and the same Mathlib as the Lean certificate of *The Chaise Longue Theorem*.

Commands, from the project root:

```
lake exe cache get          # Mathlib's compiled files
lake build                  # all 9 files; exit 0
lake env lean checks/CheckFinal.lean
                             # #check, #print of the definitions, #print axioms,
                             # and the non-vacuity example at m = 5
```

Clean rebuild. On 30 Sep 2026 the project's own build directory was moved aside (Mathlib untouched) and the 9 modules were rebuilt one at a time in dependency order, each with its own lake build inside the watchdog, so that only one Lean process ran at once. All 9 built with exit code 0, from 07:43:16 to 07:54:21 (11 min 05 s), with 658 s spent inside the 9 lake build calls: *Main* 168 s, *Defs* 40 s, *W1* 62 s, *W2* 69 s, *W3* 55 s, *W4* 63 s, *W5* 45 s, *W6* 67 s, *W7* 89 s. Afterwards a plain lake build reported «Build completed successfully (8035 jobs)». On the fresh build, checks/CheckFinal.lean printed exactly the same output as before, including the axioms [propext, Classical.choice, Quot.sound], and the non-vacuity example compiled again. Logs: clean_rebuild_2026-09-30.log, one cr_<module>.log per module, cr_all.log, checkfinal_fresh.log.

Fingerprints of the source:

- SHA-256 of the concatenation of the 9 .lean files, in sorted path order: 489fcfc254013cb8d40d41d84755887213218bb7f7a2760871850af4e6d53a0a.
- SHA-256 of the list of their individual SHA-256 lines (shasum -a 256 \$(find RequestProject -name '*.lean' | sort) | shasum -a 256): 1d331fbbf90a629ba79aee1a8d19120a9823bd77969affd6d-b7f740582cecf6e.
- SHA-256 of each file:

File	SHA-256
RequestProject/Main.lean	929b0bdeff0b781f3fb42c7a99f252dc0bda7331f698104f7075e12ff637c52d
RequestProject/Watermark/Defs.lean	0af5ba018edd1a64138a40349e3d448062cdb20dbb5c2634198f47ff19af12b5
RequestProject/Watermark/W1.lean	fe5a814f4feba342697dd66dcaa9d95d403a06d4e458afc9014e08664649075a
RequestProject/Watermark/W2.lean	29580b50a72a8b11bf02c2f622198e632ff92b0df4386ebe4fd5dc46395c9342
RequestProject/Watermark/W3.lean	61a2e26993834c755c580c39dd01b0b936f5657eeef8f88c4c4208a9ffbefbd7
RequestProject/Watermark/W4.lean	a2419a27d3bf7d6c14e94135258f51271142b7049695ed379c84e78553077a06
RequestProject/Watermark/W5.lean	05a5311a1761f00fc1c260548d0d6f43a981384abe286a59f1ec3a2d42eb4858
RequestProject/Watermark/W6.lean	f0fbd352eee1f075ec7640643fe68b0eb8303ef4404be7a4155644309b734839
RequestProject/Watermark/W7.lean	bc93a6455c604248ebfa186150b7f6eb66662c0c65c4ad1738df8d499f473a58

The project now holds 12 files. After this certificate, the three files of the Double Ladder (RequestProject/DoubleLadder/DL1.lean, DL2.lean, DL3.lean) were added to the same project [DL-Lean]. The nine files above are unchanged: their SHA-256 values, and the SHA-256 of their concatenation, were checked again on 30 September after the Double Ladder was certified, and they are identical. The fingerprints of all 12 files are in [DL-Lean, §8].

Where the files are published. The repository github.com/tretoef-estrella/watermark-theorem holds the whole Lean folder as `lean/` (see its README.md). There, the Lean project `proyecto/` is *lean/project/*, and everything else listed below is in *lean/watermark/*, with the same names and the same relative paths. The returned projects `run1/ ... run7/` are not published: each is a full copy of the project at that stage, and the final files are in `project/`. The Macaulay2 outputs `checks/kerG5.txt` and `checks/kerG7.txt` (118 MB) are not published either; `checks/kerexp5.m2` and `checks/kerexp7.m2` regenerate them. From the root of `project/`, the check of §8 is `lake env lean ../watermark/checks/CheckFinal.lean`.

On the author's machine (folder ARISTOTLE_LEAN/WATERMARK/ of the author's archive):

- `proyecto/`: the Lean project;
- `run1/ ... run7/`: every returned project, extracted;
- `pieces/`: the seven pieces sent;
- `checks/`: the brute-force scripts and their logs, the Lean check files `Check1.lean ... Check7.lean`, `CheckFinal.lean`, `DepsFinal.lean`, and the geometry check `wfinal_geometry.py`;
- `build_run1.log ... build_run7.log`, `check_run1.log ... check_run7.log`: every build and every check, for all seven runs;
- `checkfinal.log`, `checkfinal_fresh.log`, `deps_watermark.log`, `clean_rebuild_2026-09-30.log` and the per-module logs `cr_*.log`;
- `ESTADO_WATERMARK.md`: the state file, one section per run.

Recommended to a reader who wants to remove the remaining trust in the author's logs: an independent lake build on another machine, and a run of lean4checker on the compiled project. Neither has been done yet.

9. What this certificate claims, in one paragraph

A computer proof assistant, Lean 4 with the Mathlib library, has checked a complete proof of the following statement, with no gap and no axiom beyond the three standard ones:

For every prime $m \geq 5$, let G be the explicit $3m^2 \times 3m^2$ integer matrix of §1.2 and $V = Z^{3m^2}/\ker G$ with the form induced by G . Then V is a free abelian group of rank $3(m-1)(m-2)+1$, and the Gram determinant of every Z -basis of V equals $m^{3(m-3)^2}$.

G is the intersection matrix of the $3m^2$ lines on the Fermat surface S_m (§1.4: proved by hand, checked by computer, not formalized). So V is the lattice of the lines, and by [SSvL] and [Deg] it is $NS(S_m)$ for prime m . The statement is therefore the formula $\text{disc } NS(S_m) = m^{3(m-3)^2}$, asked for by Shioda in 1987 (Questions 7.2 and 7.4), for every prime $m \geq 5$.

The proofs in Lean were written by the AI system Aristotle (Harmonic) from pieces written by the author. The author's audit, assisted by Claude (Anthropic), checked that the statements match the paper, and compiled and re-checked everything on his own machine.

10. What was known before

This section records what we found in the literature on 30 September 2026. A search cannot prove that nothing exists; it records what we looked at. Every source below was read in the original.

- **Shioda (1987)** [Sh87], §7, p. 133. *Question 7.2*: for the Fermat surface of prime degree m in characteristic $p \equiv 1 \pmod{m}$, is $|\det NS(X)| = m^{3(m-3)^2}$? *Question 7.4 (i)*: for the complex Fermat surface of prime degree m , are (7.10) $|\det NS| = m^{3(m-3)^2}$ and (7.11) « NS is spanned by the classes of lines» true? He asks for the determinant only. «Added in proof» (1): the discriminant divides a power of m for every m . For every prime $m \geq 5$, the Watermark Theorem answers (7.10) for the lattice of the lines, and [Deg] answers (7.11); together they give (7.10) for NS .
- **Schütt, Shioda, van Luijk (2010)** [SSvL], arXiv source. End of §4: for every odd $m \leq 81$, the determinant of the intersection form on their basis of lines is $m^{3(m-3)^2}$, «with exponent as conjectured in [Sh87]». They prove that the lines span $NS \otimes Q$, and span NS for $m \leq 100$ with $\gcd(m, 6) = 1$.
- **Degtyarev (2015)** [Deg], arXiv source. For every $m \leq 4$ and every m with $\gcd(m, 6) = 1$, the lines generate $NS(S_m)$ over Z . The proof is topological; it gives no discriminant.
- **Shioda (2015)** [Sh15], §6.5, Conjecture 23: for m prime to 6, $\det NS(X_m) = m^{3(m-3)^2}$ appears inside a conjecture on Mordell–Weil lattices. In 2015 the formula was still used as expected, not as proved.
- **Aljovin, Movasati, Villaflor (2019)** [AMV], Table 1: the elementary divisors of the lattice of linear cycles of the Fermat surface, computed by machine (Smith normal form, one cell at a time) for $3 \leq m \leq 14$. For prime m the rows are (2,5): $1^{26} \cdot 5^{10} \cdot 25$; (2,7): $1^{48} \cdot 7^{38} \cdot 49^5$; (2,11): $1^{96} \cdot 11^{158} \cdot 121^{17}$; (2,13): $1^{120} \cdot 13^{254} \cdot 169^{23}$. The products are 5^{12} , 7^{48} , 11^{192} and 13^{300} , which are $m^{3(m-3)^2}$ in each case. [AMV] give no general formula and no proof for general m .

Therefore. Before June 2026 the formula $m^{3(m-3)^2}$ was asked for by Shioda for prime m , checked by computer for every odd $m \leq 81$, and read off by computer from the elementary divisors for $m \leq 14$. As far as we could determine, its proof for every prime $m \geq 5$ is new; it is now certified in Lean. The structure of the group, which goes beyond Shioda's question, is the Double Ladder Theorem [DL-Lean].

References

- [AMV] E. Aljovin, H. Movasati, R. Villaflor, *Integral Hodge conjecture for Fermat varieties*, J. Symbolic Comput. **95** (2019), 177–184; arXiv:1711.02628.
- [CL-Lean] R. Amichis Luengo, *The Chaise Longue Theorem — Lean Certificate*, v1, Zenodo 2026, DOI 10.5281/zenodo.23045371.
- [DL-Lean] R. Amichis Luengo, *The Double Ladder Theorem — Lean Certificate*, 30 September 2026; repository github.com/tretoef-estrella/watermark-theorem, folder *lean/double-ladder/*.
- [Deg] A. Degtyarev, *Lines generate the Picard groups of certain Fermat surfaces*, J. Number Theory **147** (2015), 454–477; doi:10.1016/j.jnt.2014.07.020; arXiv:1305.3073.
- [Sh87] T. Shioda, *Some observations on Jacobi sums*, in: Galois Representations and Arithmetic Algebraic Geometry (Kyoto 1985 / Tokyo 1986), Adv. Stud. Pure Math. **12** (1987), 119–135; doi:10.2969/aspm/01210119.
- [Sh15] T. Shioda, *Mordell–Weil lattice of higher genus fibration on a Fermat surface*, J. Math. Sci. Univ. Tokyo **22** (2015), 443–468.
- [SSvL] M. Schütt, T. Shioda, R. van Luijk, *Lines on Fermat surfaces*, J. Number Theory **130** (2010), no. 9, 1939–1963; doi:10.1016/j.jnt.2010.01.008; arXiv:0812.2377.
- R. Amichis Luengo, *The Watermark Theorem*, v1.0, 4 June 2026, repository github.com/tretoef-estrella/chaise-longue-theorem, [hodge-fermat-campaign](https://github.com/tretoef-estrella/hodge-fermat-campaign)/THE_WATERMARK_THEOREM.md.
- The Lean 4 theorem prover, <https://lean-lang.org>; the Mathlib library, <https://github.com/leanprover-community/mathlib4>.
- Aristotle, Harmonic, <https://aristotle.harmonic.fun>.